Food Waste Reduction Mobile Application: Real-Time Surplus Redistribution Platform

Luna A. Sandoval – Project Lead & Systems Analyst | Cristian C. Bonilla – Frontend Developer & Systems Analyst | Nicolás Rodríguez – Backend Developer & Systems Analyst | Juan S. Bravo – Database Specialist & Systems Analyst
Universidad Distrital Francisco José de Caldas - System Engineering Program – Systems Analysis Course, 2026-I

Introduction

9.7M Tons Wasted/Year

Bogotá Colombia

Cafeterias & Restaurants

In Colombia, 9.7 million tons of food are wasted annually. Specifically in Bogotá, cafeterias and local restaurants generate daily surpluses suitable for consumption, but the lack of rapid coordination and the short shelf life lead to their loss, exacerbating environmental damage, food insecurity, and social inequality. Platforms such as OLIO and EatCloud have demonstrated the potential of mobile applications; however, none are optimized for accessible environments with a strict 3 km walking radius, same-day operation, and mandatory priority for verified foundations.

For this reason, this project proposes a mobile platform specifically designed for the area surrounding the Faculty of Engineering at the Universidad Distrital Francisco José de Caldas, aimed at addressing these limitations through precise geolocation, proximity-based matching with priority for charities, and real-time notifications.

Goal

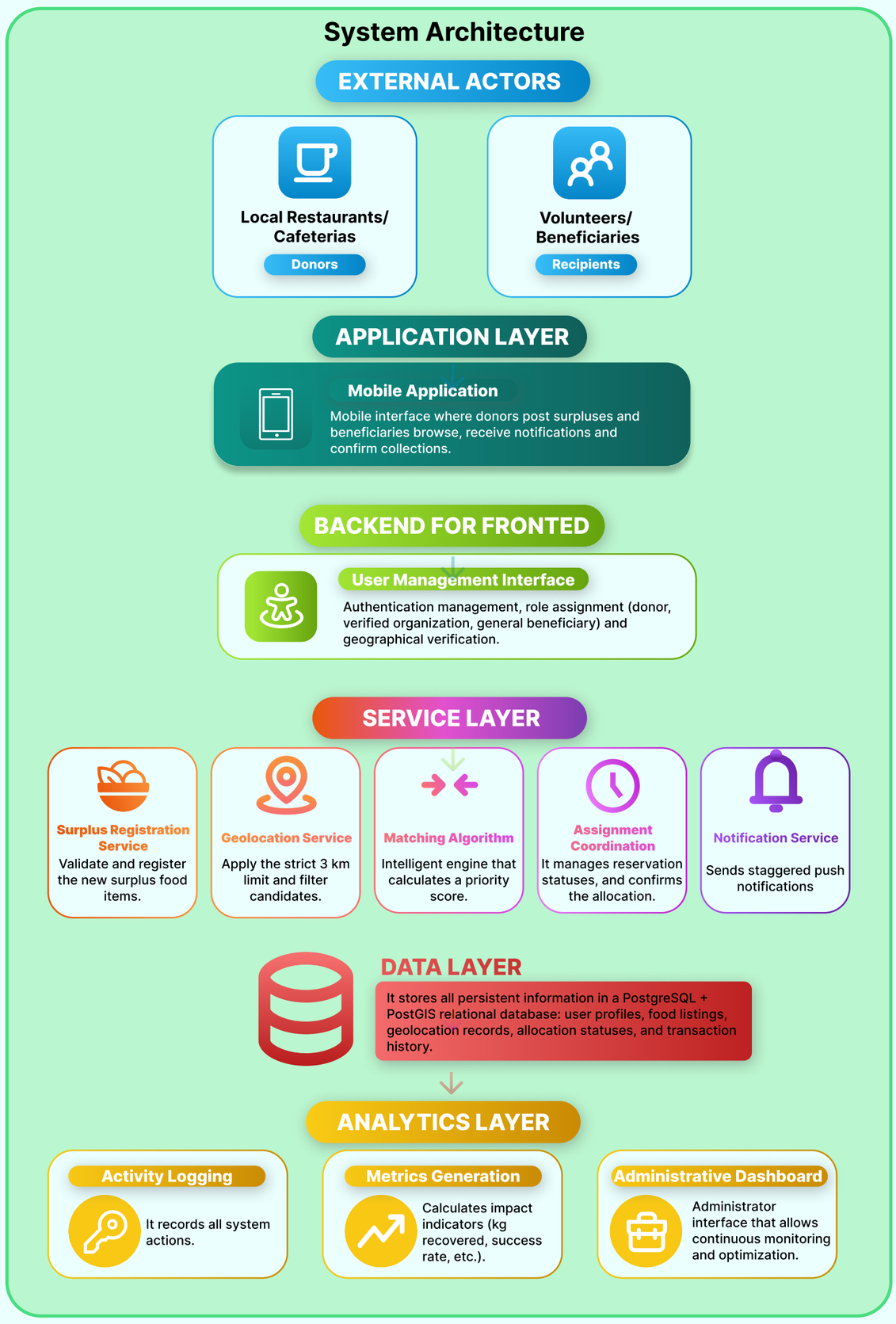
Research Question

How can a mobile platform, constrained to a 3 km walking-access radius, enable efficient real-time redistribution of food surpluses through geolocation and priority-based matching, in order to reduce food waste and improve equitable access?

Expected Product

A functional mobile platform prototype that enables real-time redistribution of food surpluses within a 3 km walking-access radius, integrating geolocation services, a proximity-based matching algorithm with priority rules for verified organizations, and a real-time notification system. The product is expected to demonstrate measurable improvements in surplus recovery rate, matching efficiency, and successful pickup rates under concurrent user scenarios.

Proposed Solution



The Food Waste Reduction Platform uses a clean four-layer architecture to enable real-time surplus food redistribution within a 3 km radius of the Engineering Faculty at Universidad Distrital Francisco José de Caldas.
External actors (restaurants/cafeterias as donors and volunters/beneficers as beneficiaries) interact through a React Native mobile application. The Service Layer processes surplus registration, enforces geospatial constraints, runs a priority-based matching algorithm, coordinates assignments, and sends tiered push notifications. All data is persisted in PostgreSQL with PostGIS, while an Analytics Layer generates impact metrics and an administrative dashboard.
Strict operational boundaries—same-day publication (2-4 h before closing) and pickup (1-3 h)—ensure food safety and logistical feasibility. This design prioritizes verified charitable organizations and delivers measurable reductions in food waste through rapid digital coordination.

Workshop 3 & 4

Experiments

Results

Conclusion

References

[1] Food Waste Reduction Mobile Application, A Rigorous System Analysis (Workshop 1). Universidad Distrital Francisco José de Caldas, Systems Analysis Course, 2026.
[2] Solution Design and Initial Development (Workshop 2). Universidad Distrital Francisco José de Caldas, Systems Analysis Course, 2026.
[3] United Nations Environment Programme (UNEP). Food Waste Index Report 2021. UNEP, 2021.
[4] Alcaldía de Bogotá. “Yo Contribuyo, No Desperdicio” – Estrategia para reducir la pérdida y desperdicio de alimentos. Secretaría Distrital de Desarrollo Económico, 2024.